Data Visualization & Analytics Lab

Experiment No. : 2

Build advanced visualizations using Tableau/Power BI

Experiment No.2

1. **Aim:** Build advanced visualizations (stacked histograms, box plots, heatmaps, bubble chart and scatter plot matrices) using Tableau/Power BI and compare how multivariate data is represented.
2. **Objectives:**
    - Understand why visualization is an important part of data analysis.
    - Understand the components involved in visualization design.

3. **Course Outcomes:** Students will understand and gain proficiency in advanced chart types for complex datasets.
4. **Hardware / Software Required: Tableau/PowerBI**

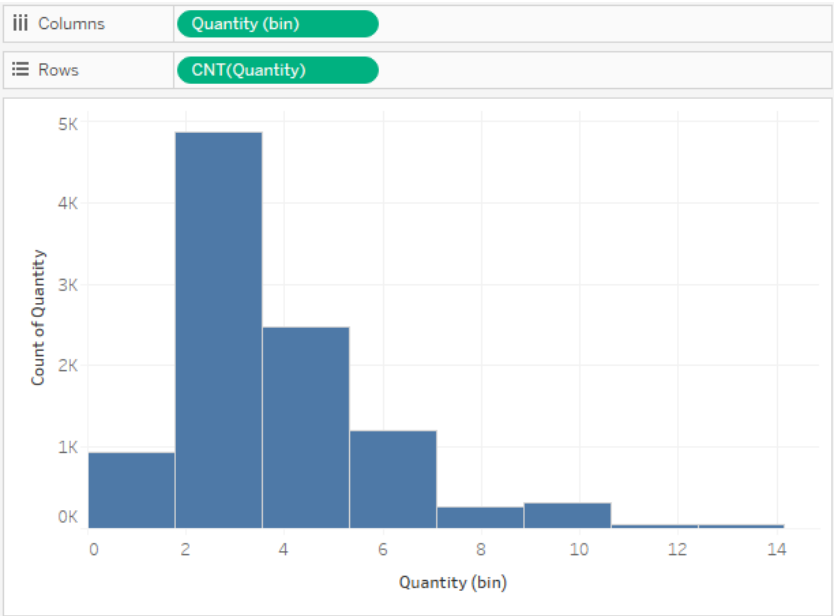
5. **Theory:** Logistic regression is a statistical machine learning algorithm that classifies the data by considering outcome variables on extreme ends and tries makes a logarithmic line that distinguishes between them.Logistic regression is another powerful supervised ML algorithm used for binary classification problems (when target is categorical). The best way to think about logistic regression is that it is a linear regression but for classification problems. Logistic regression essentially uses a logistic function defined below to model a binary output variable. The primary difference between linear regression and logistic regression is that logistic regression's range is bounded between 0 and 1.

1. Histograms

- **Purpose**: A histogram is used to visualize the distribution of a single variable. It divides data into bins (intervals) and shows the frequency of data points in each bin.

Tableau:

- **Steps**:
    1. Drag a field (e.g., quantity, sales, age, etc.) to the Columns shelf.
    2. Change the chart type to "Histogram" from the Show Me menu.
    3. Tableau automatically groups data into bins and plots the frequency distribution.
- **Customization**:
    - You can adjust bin sizes or the axis range.
    - Add color encoding for additional insights like grouping by categories.



Three things happen after you click the histogram icon in **Show Me**:

- The view changes to show vertical bars, with a continuous x-axis (1 – 14) and a continuous y-axis (0 – 5,000).
- The **Quantity** measure you placed on the **Columns** shelf, which had been aggregated as SUM, is replaced by a continuous **Quantity (bin)** dimension. (The green color of the field on the **Columns** shelf indicates that the field is continuous.)
- The **Quantity** measure moves to the **Rows** shelf and the aggregation changes from SUM to CNT (Count).

2. Box Plots

- **Purpose**: Box plots , also known as box-and-whisker plots , show the spread of data, including the median, quartiles, and outliers.
- Boxes indicate the middle 50 percent of the data (that is, the middle two quartiles of the data's distribution). You can configure lines, called *whiskers*, to display all points within 1.5 times the interquartile range (in other words, all points within 1.5 times the width of the adjoining box), or all points at the maximum extent of the data, as shown in the following image:
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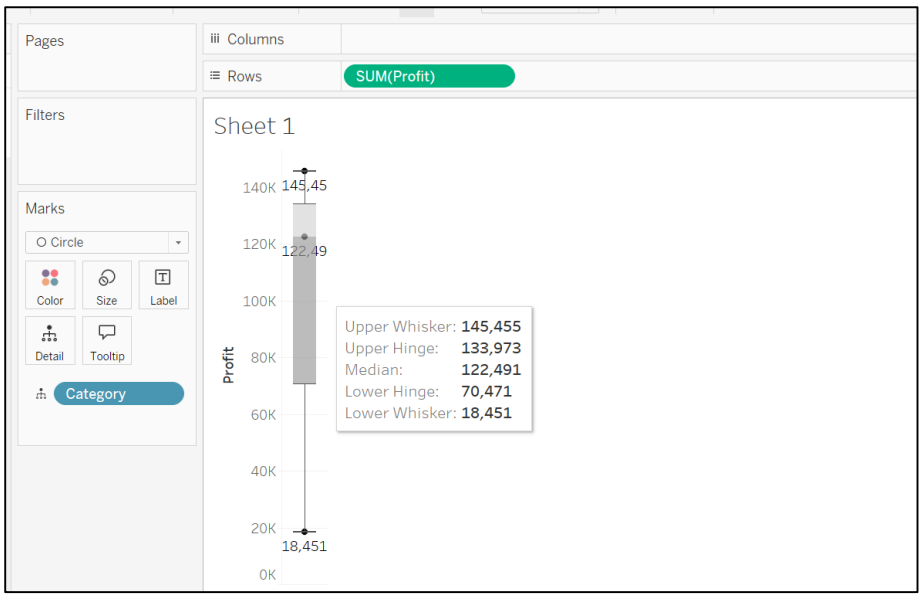
Tableau:

- **Steps**:

1. Drag a measure (e.g., sales) to Rows and a dimension (e.g., category) to Columns.
2. Select “Box Plot” from the Show Me panel.

- **Customization**:
  - Tableau allows customization of the whiskers, outlier definitions, and color coding.



Power BI:

- **Steps**:
  1. Add a Box and Whisker chart from the Visualizations pane (again, may require a custom visual from the marketplace).
  2. Add the desired measure and dimension to the respective axes.
- **Customization**:
  - Custom visual settings allow tweaking of how outliers and quartiles are displayed.

Comparison:

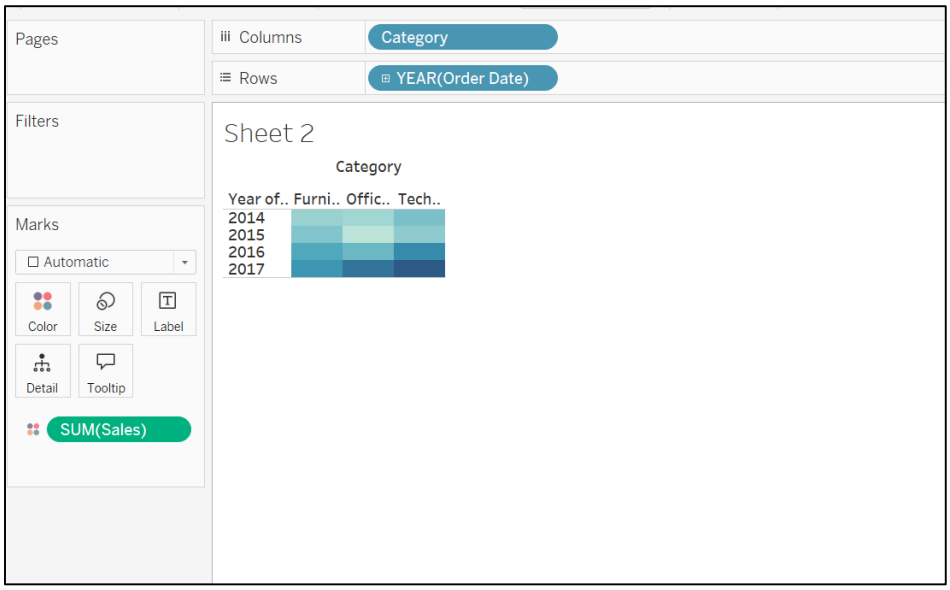
- **Tableau** has built-in box plot support, offering straightforward configuration and in-depth customization.
- **Power BI** requires custom visuals for box plots, but it provides a similar level of customization once the visual is added.

3. Heatmaps

- **Purpose**: A heatmap visualizes data through color gradients. It is particularly useful for visualizing correlations and patterns between two variables across a grid of dimensions.

Tableau:

- **Steps**:
  1. Drag two dimensions (e.g., product categories, time periods) to Columns and Rows.
  2. Drag a measure (e.g., sales) to Color.
  3. Tableau will automatically generate a heatmap with color gradients.
- **Customization**:
  - You can change the color palette, adjust the cell size, and add labels for better insight.



Power BI:

- **Steps**:
  1. Use a matrix visualization and drag two dimensions to Rows and Columns.
  2. Drag a measure to Values, and then format the values as a heatmap using conditional formatting.
- **Customization**:

- Power BI's heatmap functionality is achieved by using conditional formatting (color scales) to visually represent the measure.

Comparison:

- **Tableau** offers a more traditional heatmap visualization that integrates smoothly with its drag-and-drop interface.
- **Power BI** relies on conditional formatting within a matrix visualization, which is versatile but requires more steps for advanced features.

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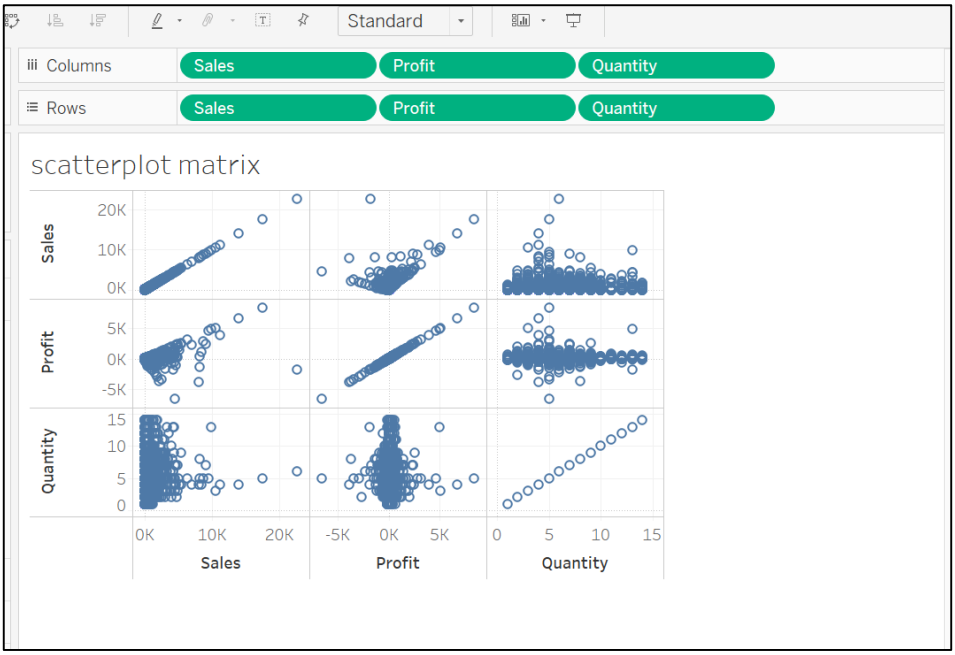
4. Scatter Plot Matrices

- **Purpose**: A scatter plot matrix (or pair plot) visualizes the relationships between multiple numerical variables in a dataset by showing scatter plots of pairs of variables.

Tableau:

- **Steps**:
  1. Drag multiple measures (e.g., age, income, and spending) to the Columns and Rows.
  2. Use the "Pair Plot" option or create a scatter plot for each pair manually.
- **Customization**:
  - You can easily adjust the colors, size, and shape of the data points to represent additional dimensions.



Power BI:

- **Steps**:
  1. Use a scatter plot for each pair of variables, or use custom visuals that enable scatter plot matrices (like the "Pair Plot" visual).
  2. Power BI allows you to layer the data points with other dimensions for deeper insights.
- **Customization**:
  - Power BI's customization is more focused on individual scatter plots, so creating a full matrix would require adding multiple visuals.

Comparison:

- **Tableau** excels in creating interactive scatter plot matrices with a lot of flexibility for exploration.
- **Power BI** can generate similar scatter plots but lacks the built-in matrix functionality, requiring manual setup for full scatter plot matrix views.

6. **Conclusion:** We have learnt about advanced data visualization techniques like box plot, heat map, pair plot etc.

8. Viva Questions:

- What are the advanced visualization techniques?
- What visual elements are suitable for numerical and categorical data??

References:

- https://www.tableau.com/chart
- https://learn.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-types-for-reports-and-q-and-a